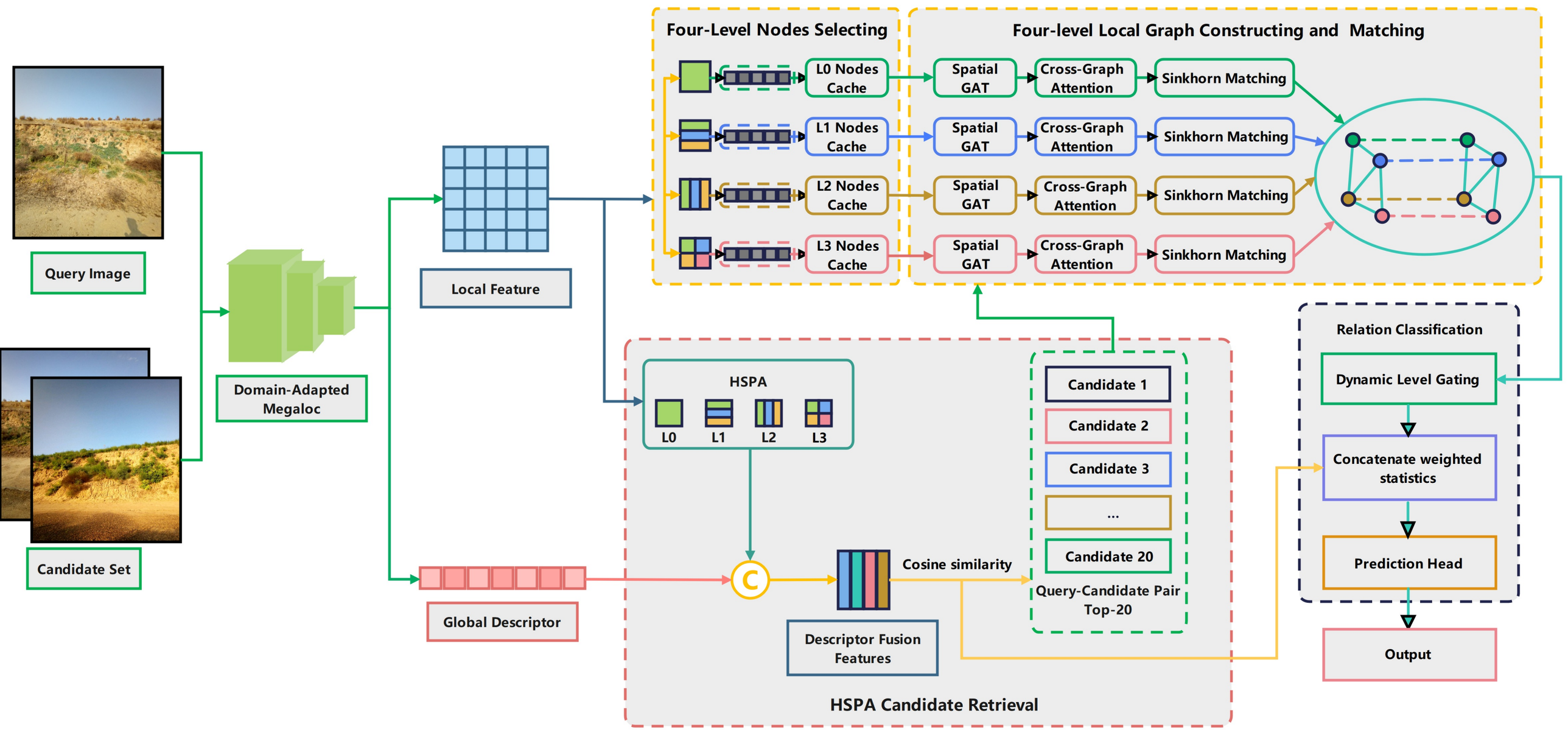

Four-Level Nodes Selecting
Four-level Local Graph Constructing and Matching
Query Image
Candidate Set
Domain-Adapted Megaloc
Local Feature
Global Descriptor
L0 Nodes Cache
L1 Nodes Cache
L2 Nodes Cache
L3 Nodes Cache
Spatial GAT
Cross-Graph Attention
Sinkhorn Matching
HSPA
L0
L1
L2
L3
C
Descriptor Fusion Features
Cosine similarity
Candidate 1
Candidate 2
Candidate 3
...
Candidate 20
Query-Candidate Pair Top-20
HSPA Candidate Retrieval
Relation Classification
Dynamic Level Gating
Concatenate weighted statistics
Prediction Head
Output